Assign 3 : Linux Commands

Setup (working directory)

```
rahulastic@IDEAPAD:~$ mkdir assign3
rahulastic@IDEAPAD:~$ cd assign3
```

1. ls – Display hidden files

```
rahulastic@IDEAPAD:~/assign3$ touch .hidden1 .hidden2 file1.txt file2.txt file3.txt
rahulastic@IDEAPAD:~/assign3$ ls
file1.txt  file2.txt  file3.txt
rahulastic@IDEAPAD:~/assign3$ ls -a
.  ..  .hidden1  .hidden2  file1.txt  file2.txt  file3.txt
```

2. rename – Rename all *.txt files to *.htm

```
rahulastic@IDEAPAD:~/assign3$ rename 's/\.txt$/.htm/' *.txt
rahulastic@IDEAPAD:~/assign3$ ls
file1.htm  file2.htm  file3.htm
```

3. ln -s – Create symbolic link (shortcut) for a file

```
rahulastic@IDEAPAD:~/assign3$ cat > hello.txt
Hello Linux
Welcome to Linux world
Linux commands are easy
Learning is fun
rahulastic@IDEAPAD:~/assign3$ ln -s hello.txt link_hello
rahulastic@IDEAPAD:~/assign3$ ls -l
total 4
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:00 file1.htm
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:00 file2.htm
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:00 file3.htm
-rw-r--r-- 1 rahulastic rahulastic 75 Oct 28 12:04 hello.txt
lrwxrwxrwx 1 rahulastic rahulastic  9 Oct 28 12:05 link_hello -> hello.txt
rahulastic@IDEAPAD:~/assign3$ cat link_hello
Hello Linux
Welcome to Linux world
Linux commands are easy
Learning is fun
```

4. ln – Create hard link for a file

```
rahulastic@IDEAPAD:~/assign3$ ln hello.txt hard_hello
rahulastic@IDEAPAD:~/assign3$ ls -li hello.txt hard_hello
524801 -rw-r--r-- 2 rahulastic rahulastic 75 Oct 28 12:04 hard_hello
524801 -rw-r--r-- 2 rahulastic rahulastic 75 Oct 28 12:04 hello.txt
```

5. stat – Display time stamp of the file

```
rahulastic@IDEAPAD:~/assign3$ stat hello.txt
  File: hello.txt
  Size: 75          Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d  Inode: 524801     Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/rahulastic)   Gid: ( 1000/rahulastic)
Access: 2026-10-28 12:05:41.482913700 +0530
Modify: 2026-10-28 12:04:35.119204500 +0530
Change: 2026-10-28 12:06:02.730118200 +0530
 Birth: 2026-10-28 12:04:35.119204500 +0530
```

6. touch – Change the time stamp of the file

```
rahulastic@IDEAPAD:~/assign3$ touch hello.txt
rahulastic@IDEAPAD:~/assign3$ stat hello.txt | grep Modify
Modify: 2026-10-28 12:10:44.213507000 +0530
rahulastic@IDEAPAD:~/assign3$ touch -t 202610011030 hello.txt
rahulastic@IDEAPAD:~/assign3$ stat hello.txt | grep Modify
Modify: 2026-10-01 10:30:00.000000000 +0530
```

7. touch – Create empty file

```
rahulastic@IDEAPAD:~/assign3$ touch empty.txt
rahulastic@IDEAPAD:~/assign3$ ls -l empty.txt
-rw-r--r-- 1 rahulastic rahulastic 0 Oct 28 12:12 empty.txt
```

8. tail – Print all lines in a file from fifth line onwards

```
rahulastic@IDEAPAD:~/assign3$ seq 1 12 > lines.txt
rahulastic@IDEAPAD:~/assign3$ tail -n +5 lines.txt
5
6
7
8
9
10
11
12
```

9. init 0, poweroff – Shutdown (only for super-user)

Using init:

```
rahulastic@IDEAPAD:~/assign3$ sudo init 0
[sudo] password for rahulastic:
```

Using poweroff:

```
rahulastic@IDEAPAD:~/assign3$ sudo poweroff
[sudo] password for rahulastic:
```

Note: these commands shut the system down and need super-user rights, so they are shown here but not run.

10. init 6, reboot – Reboot (only for super-user)

Using init:

```
rahulastic@IDEAPAD:~/assign3$ sudo init 6
[sudo] password for rahulastic:
```

Using reboot:

```
rahulastic@IDEAPAD:~/assign3$ sudo reboot
[sudo] password for rahulastic:
```

Note: these commands restart the system and need super-user rights, so they are shown here but not run.

12. whoami – Find current user

```
rahulastic@IDEAPAD:~/assign3$ whoami
rahulastic
```

13. tty – Find current terminal

```
rahulastic@IDEAPAD:~/assign3$ tty
/dev/pts/0
```

14. who am i – Find current user and terminal

```
rahulastic@IDEAPAD:~/assign3$ who am i
rahulastic pts/0        2026-10-28 11:58
```

15. users – Find all users logged into the system

```
rahulastic@IDEAPAD:~/assign3$ users
amit rahulastic rahulastic rohan sneha
```

16. who – Find all users logged in and terminals in use

```
rahulastic@IDEAPAD:~/assign3$ who
rahulastic tty1         2026-10-28 09:15
amit     tty2         2026-10-28 09:40
sneha    tty3         2026-10-28 10:05
rohan    tty4         2026-10-28 10:20
rahulastic pts/0        2026-10-28 11:58
```

17. grep, egrep, fgrep – Find a given word/pattern from the file

```
rahulastic@IDEAPAD:~/assign3$ grep Linux hello.txt
Hello Linux
Welcome to Linux world
Linux commands are easy
rahulastic@IDEAPAD:~/assign3$ grep -i learning hello.txt
Learning is fun
rahulastic@IDEAPAD:~/assign3$ egrep "Hello|fun" hello.txt
egrep: warning: egrep is obsolescent; using grep -E
Hello Linux
Learning is fun
rahulastic@IDEAPAD:~/assign3$ fgrep "Linux" hello.txt
fgrep: warning: fgrep is obsolescent; using grep -F
Hello Linux
Welcome to Linux world
Linux commands are easy
```

18. chmod – Make a file as read-only

```
rahulastic@IDEAPAD:~/assign3$ chmod 444 hello.txt
rahulastic@IDEAPAD:~/assign3$ ls -l hello.txt
-r--r--r-- 2 rahulastic rahulastic 75 Oct  1 10:30 hello.txt
rahulastic@IDEAPAD:~/assign3$ echo test >> hello.txt
-bash: hello.txt: Permission denied
```

19. chmod – Make a file as write, read and execute

```
rahulastic@IDEAPAD:~/assign3$ chmod 777 hello.txt
rahulastic@IDEAPAD:~/assign3$ ls -l hello.txt
-rwxrwxrwx 2 rahulastic rahulastic 75 Oct  1 10:30 hello.txt
```

20. wc (using redirection) – Take input from the file, count words and store output to other file

```
rahulastic@IDEAPAD:~/assign3$ wc < hello.txt > count.txt
rahulastic@IDEAPAD:~/assign3$ cat count.txt
 4 13 75
```

21. who | wc (using pipe) – Count words in output of "who" command

```
rahulastic@IDEAPAD:~/assign3$ who | wc -w
20
rahulastic@IDEAPAD:~/assign3$ who | wc -l
5
```

22. head, tail (using pipe) – Print lines 5 to 10 from the given file

```
rahulastic@IDEAPAD:~/assign3$ head -n 10 lines.txt | tail -n 6
5
6
7
8
9
10
```

23. sort, uniq (using pipe) – Print the unique values from the unsorted file

```
rahulastic@IDEAPAD:~/assign3$ cat > names.txt
banana
apple
banana
mango
apple
cherry
rahulastic@IDEAPAD:~/assign3$ sort names.txt | uniq
apple
banana
cherry
mango
```

24. who, grep (using pipe) – Display all users not using tty2, tty3, tty4

```
rahulastic@IDEAPAD:~/assign3$ who | grep -v -E 'tty2|tty3|tty4'
rahulastic tty1         2026-10-28 09:15
rahulastic pts/0        2026-10-28 11:58
```

25. ls – Display all files in ascending order of their file size

```
rahulastic@IDEAPAD:~/assign3$ ls -lSr
total 20
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:00 file3.htm
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:00 file2.htm
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:00 file1.htm
-rw-r--r-- 1 rahulastic rahulastic  0 Oct 28 12:12 empty.txt
lrwxrwxrwx 1 rahulastic rahulastic  9 Oct 28 12:05 link_hello -> hello.txt
-rw-r--r-- 1 rahulastic rahulastic  9 Oct 28 12:20 count.txt
-rw-r--r-- 1 rahulastic rahulastic 27 Oct 28 12:15 lines.txt
-rw-r--r-- 1 rahulastic rahulastic 39 Oct 28 12:25 names.txt
-rwxrwxrwx 2 rahulastic rahulastic 75 Oct  1 10:30 hello.txt
-rwxrwxrwx 2 rahulastic rahulastic 75 Oct  1 10:30 hard_hello
```

26. ls, grep (using pipe) – Count number of files and directories from current directory

```
rahulastic@IDEAPAD:~/assign3$ mkdir docs projects
rahulastic@IDEAPAD:~/assign3$ ls -l | grep '^-' | wc -l
9
rahulastic@IDEAPAD:~/assign3$ ls -l | grep '^d' | wc -l
2
rahulastic@IDEAPAD:~/assign3$ ls | wc -l
12
```

27. ls, grep (using pipe) – Display only hidden files in a directory

```
rahulastic@IDEAPAD:~/assign3$ ls -a | grep '^\.'
.
..
.hidden1
.hidden2
rahulastic@IDEAPAD:~/assign3$ ls -A | grep '^\.'
.hidden1
.hidden2
```

28. tr – Convert file contents into uppercase, also learn "-s" option

```
rahulastic@IDEAPAD:~/assign3$ tr 'a-z' 'A-Z' < hello.txt
HELLO LINUX
WELCOME TO LINUX WORLD
LINUX COMMANDS ARE EASY
LEARNING IS FUN
rahulastic@IDEAPAD:~/assign3$ echo "hello     world    linux" | tr -s " "
hello world linux
rahulastic@IDEAPAD:~/assign3$ echo "aabbccdd" | tr -s "a-d"
abcd
```

29. cut – Split the file contents using delimiter and print required fields

```
rahulastic@IDEAPAD:~/assign3$ cat > students.csv
101,Amit,Pune,85
102,Sneha,Mumbai,92
103,Rohan,Nashik,78
rahulastic@IDEAPAD:~/assign3$ cut -d, -f2 students.csv
Amit
Sneha
Rohan
rahulastic@IDEAPAD:~/assign3$ cut -d, -f1,3 students.csv
101,Pune
102,Mumbai
103,Nashik
rahulastic@IDEAPAD:~/assign3$ cut -c1-3 students.csv
101
102
103
```